

Health & Safety

Standard Dangerous energies sources



Reference

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1. Objective

The purpose of this document is to describe the types of dangerous energies that may be encountered during on-site operations and to specify in which documentation Bouygues Travaux Publics (BYTP) requirements are defined to prevent the associated risks.

This involves preventing the risks associated with these types of activities in compliance with the regulations, codes and standards in force in all the countries where BYTP operates.

2. Scope

This document applies to all companies working on sites (including service providers and subcontractors).

In the case of Joint ventures (JV), Consortiums or co-contracting operations, the project-specific procedure must include, as a minimum, the provisions set out in this document as well as any non-conflicting provisions of our partners. It is important to have all aspects adopted at the start of the project either prior to or as a minimum at Health and Safety kick-off meetings.

If any part of this document conflicts with a local requirement and is less stringent than that local regulation, then the corresponding local regulation(s) apply.

3. Reference documents

- BYCN guidelines Hazardous energies
- BYTP-H&S-REF-2200 Standard Lines of defence and Life Saving Rules
- BYTP-H&S-REF-2212 Standard Vehicles pedestrian interaction
- BYTP-H&S-REF-2215 Standard Construction machinery and associated maintenance
- BYTP-H&S-REF-2216 Standard Site installations
- BYTP-H&S-REF-2220 Standard portable tools
- BYTP-H&S-REF-2222 Standard Hazardous chemicals
- BYTP-H&S-REF-2224 Standard Work with high pressure tools
- BYTP-H&S-REF-2230 Standard Work with concrete
- BYTP-H&S-REF-2231 Standard Lock-out tag-out
- BYTP-H&S-REF-2232 Standard Radiation protection for Civil Works activities
- Policy regarding sensitive equipment BYTP and Subsidiaries

4. Definitions and abbreviations

Term / Abbreviation	Description
Source of dangerous energies	Any source of electrical, mechanical, hydraulic, pneumatic, chemical, thermal, radioactive or other energy where the uncontrolled exposition can cause, immediately or in the long term, serious bodily harm or severe deterioration of the health.

5. Energy sources and associated requirements

5.1. Electrical energies

The risks for humans linked to electricity are of different natures. These mainly concern the risks of electrification, electrocution and burns. These risks originate from direct or indirect contact and electric arcs.

The requirements relating to electrical installations on site and in particular in terms of verifications of electrical installations are defined in the **Standard Site installations (BYTP-H&S-REF-2216)**.

When working in an electrical environment, it is required to operate in compliance with local regulations and according to the LOTO requirements defined in the **Standard Lock-out tag-out (BYTP-H&S-REF-2231)**.

The requirements related to the management of interfaces with existing networks, buried or aerial, are defined in the **Standard Construction machinery and associated maintenance (BYTP-H&S-REF-2215)**.

5.2. Mechanical energies

Mechanical energy is the energy contained in an object under tension (cable, spring, piston, etc.) and the energy of an equipment or a moving part of the equipment (grinder, construction machinery , etc.). Contact with mechanical energy may cause serious injury by crushing, impact, etc.

It is therefore necessary to ensure that this equipment is used according to the requirements defined in the standards:

- **Standard Vehicles pedestrian interaction (BYTP-H&S-REF-2212)**
- **Standard portable tools (BYTP-H&S-REF-2220)**

When working on live equipment and when feasible and necessary, the mechanical energy lock-out steps are carried out according to the LOTO requirements defined in the **Standard Lock-out tag-out (BYTP-H&S-REF-2231)**.

5.3. Hydraulic and pneumatic energies

The risks for humans linked to hydraulic and pneumatic energies are of different natures. These mainly concern risks linked to exposure conditions (composition, temperature, quantity, pressure, etc.) which can lead to serious injuries.

It is therefore necessary to ensure that activities involving pressure equipment (high pressure cleaning, hydrodemolition, sprayed concrete, tire inflation, etc.) according to the requirements defined in the standards:

- **Standard Work with high pressure tools (BYTP-H&S-REF-2224)**
- **Standard Work with concrete (BYTP-H&S-REF-2230)**

During intervention operations on hydraulic or pneumatic networks, it is necessary to operate in compliance with local regulations and according to the lockout requirements defined in the **Standard Lock-out tag-out (BYTP-H&S-REF-2231)**.

5.4. Chemical energies

Chemical products may be hazardous for people, for the installations or for the environment: acute poisoning, asphyxiation, fire, explosion, pollution.

It is necessary to ensure the storage, use and disposal of chemical waste in accordance with local regulations and according to the requirements defined in the **Standard Hazardous chemicals (BYTP-H&S-REF-2222)** .

When working on networks containing or which may contain chemical products, it is necessary to operate in compliance with local regulations and according to the LOTO requirements defined in the **Standard Lock-out tag-out (BYTP-H&S-REF-2231)**.

5.5. Radioactive energies

Radioactive energies have acute health effects, such as skin burns or acute radiation syndrome, which can be seen when radiation doses exceed very high levels. Low doses of ionizing radiation can increase the risk of long-term effects, such as the development of cancer.

As soon as operations are carried out in a nuclear energy environment (Nuclear plant, etc.) or as soon as it is necessary to use a radioactive source on site (use of radioisotopes within the scope of Non-Destructive Testing, etc.), it is necessary to intervene in accordance with local regulations and according to the requirements defined in the **Standard Radiation protection for Civil Works activities (BYTP-H&S-REF-2232)**.

5.6. Extreme temperature energies

Extreme thermal energies are varied and present risks to people such as burns caused by exposure to high temperatures and frostbite resulting from exposure to very low temperatures. These risks arise from the presence of very hot surfaces (welding, thermal cutting, bitumen, boilers, pipelines, etc.), industrial heating equipment or refrigeration installations, as well as during the handling of hot or cryogenic fluids.

When carrying out work on equipment or systems that may present high or low temperatures, it is necessary to operate in accordance with local regulations and according to the isolation requirements defined in the **Standard Lock-out tag-out (BYTP-H&S-REF-2231)**.

Any intervention on equipment or installations presenting extreme thermal energies must be subject to a risk assessment, which should be presented to the teams during the start-of-shift briefing, and, if necessary, appropriate protective equipment should be used (heat-resistant gloves, insulated clothing, etc.) in accordance with the **Standard PPE (BYTP-H&S-REF-2219)**.